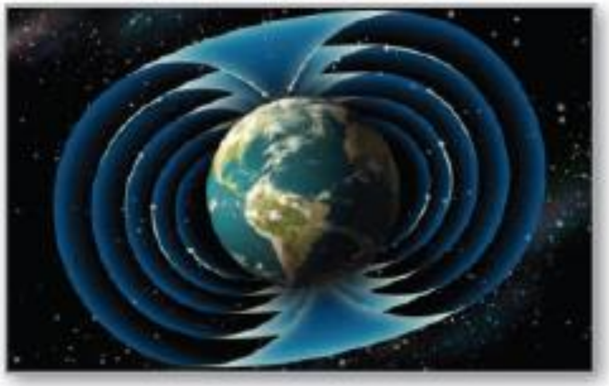
15 Vector Analysis



15.2 Line Integrals

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Objectives

- Understand and use the concept of a piecewise smooth curve.
- Write and evaluate a line integral.
- Write and evaluate a line integral of a vector field.
- Write and evaluate a line integral in differential form.

Piecewise Smooth Curves

Piecewise Smooth Curves

A classic property of gravitational fields is that, subject to certain physical constraints, the work done by gravity on an object moving between two points in the field is independent of the path taken by the object.

One of the constraints is that the **path** must be a piecewise smooth curve. We know that a plane curve *C* given by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}, \qquad a \leq t \leq b$$

is **smooth** when

$$\frac{dx}{dt} \quad \text{and} \quad \frac{dy}{dt}$$

are continuous on [*a*, *b*] and not simultaneously 0 on (*a*, *b*).

Piecewise Smooth Curves

Similarly, a space curve *C* given by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}, \quad a \le t \le b$$

is **smooth** when

$$\frac{dx}{dt}, \quad \frac{dy}{dt}, \quad \text{and} \quad \frac{dz}{dt}$$

are continuous on [*a*, *b*] and not simultaneously 0 on (*a*, *b*).

A curve *C* is **piecewise smooth** when the interval [*a*, *b*] can be partitioned into a finite number of subintervals, on each of which *C* is smooth.

Example 1 – *Finding a Piecewise Smooth Parametrization*

Find a piecewise smooth parametrization of the graph of *C* shown in Figure 15.7.

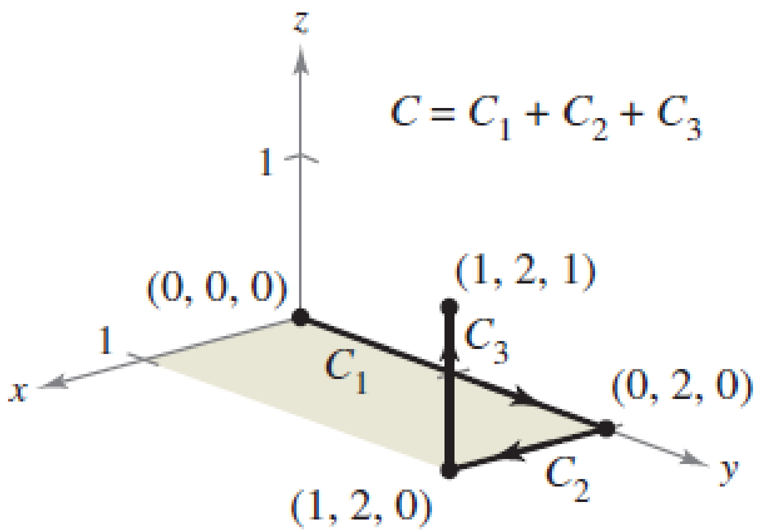


Figure 15.7

Example 1 – *Solution*

Because C consists of three line segments C_1, C_2, and C_3, you can construct a smooth parametrization for each segment and piece them together by making the last t-value in C_i correspond to the first t-value in C_{i+1}.

$$C_1\colon\ x(t) = 0, \qquad y(t) = 2t, \qquad z(t) = 0, \qquad 0 \le t \le 1$$

$$C_2\colon\ x(t) = t - 1, \qquad y(t) = 2, \qquad z(t) = 0, \qquad 1 \le t \le 2$$

$$C_3\colon\ x(t) = 1, \qquad y(t) = 2, \qquad z(t) = t - 2, \qquad 2 \le t \le 3$$

Example 1 – *Solution*

cont'd

So, C is given by

$$\mathbf{r}(t) = \begin{cases} 2t\mathbf{j}, & 0 \le t \le 1 \\ (t-1)\mathbf{i} + 2\mathbf{j}, & 1 \le t \le 2 \\ \mathbf{i} + 2\mathbf{j} + (t-2)\mathbf{k}, & 2 \le t \le 3 \end{cases}.$$

Because C_1, C_2, and C_3 are smooth, it follows that C is piecewise smooth.

Piecewise Smooth Curves

Parametrization of a curve induces an **orientation** to the curve.

For instance, in Example 1, the curve is oriented such that the positive direction is from (0, 0, 0), following the curve to (1, 2, 1).

Line Integrals

Line Integrals

In this section, you will study a new type of integral called a **line integral**

$$\int_C f(x, y)\, ds \qquad \text{Integrate over curve } C.$$

for which you integrate over a piecewise smooth curve *C*.

To introduce the concept of a line integral, consider the mass of a wire of finite length, given by a curve *C* in space.

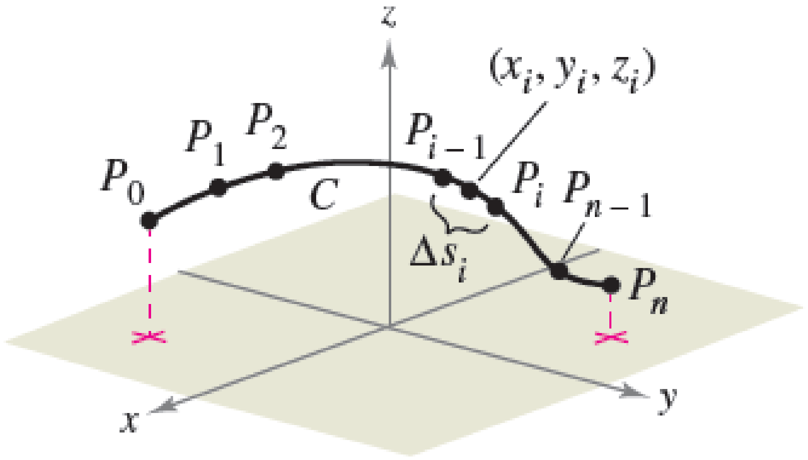
The density (mass per unit length) of the wire at the point (x, y, z) is given by $f(x, y, z)$.

Line Integrals

Partition the curve C by the points

$$P_0, P_1, \ldots, P_n$$

producing n subarcs, as shown in Figure 15.8.



Partitioning of curve C

Figure 15.8

Line Integrals

The length of the *i*th subarc is given by Δs_i.

Next, choose a point (x_i, y_i, z_i) in each subarc.

If the length of each subarc is small, then the total mass of the wire can be approximated by the sum

$$\text{Mass of wire} \approx \sum_{i=1}^{n} f(x_i, y_i, z_i)\, \Delta s_i.$$

By letting $\|\Delta\|$ denote the length of the longest subarc and letting $\|\Delta\|$ approach 0, it seems reasonable that the limit of this sum approaches the mass of the wire.

Line Integrals

Definition of Line Integral

If f is defined in a region containing a smooth curve C of finite length, then the **line integral of f along C** is given by

$$\int_C f(x, y)\, ds = \lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(x_i, y_i)\, \Delta s_i \qquad \text{Plane}$$

or

$$\int_C f(x, y, z)\, ds = \lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(x_i, y_i, z_i)\, \Delta s_i \qquad \text{Space}$$

provided this limit exists.

Line Integrals

To evaluate a line integral over a plane curve *C* given by $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$, use the fact that

$$ds = \|\mathbf{r}'(t)\|\, dt = \sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt.$$

A similar formula holds for a space curve, as indicated in Theorem 15.4.

THEOREM 15.4 Evaluation of a Line Integral as a Definite Integral

Let f be continuous in a region containing a smooth curve C. If C is given by $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$, where $a \le t \le b$, then

$$\int_C f(x, y)\, ds = \int_a^b f(x(t), y(t))\sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt.$$

If C is given by $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, where $a \le t \le b$, then

$$\int_C f(x, y, z)\, ds = \int_a^b f(x(t), y(t), z(t))\sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\, dt.$$

Line Integrals

Note that if $f(x, y, z) = 1$, then the line integral gives the arc length of the curve C. That is,

$$\int_C 1\, ds = \int_a^b \|\mathbf{r}'(t)\|\, dt = \text{length of curve } C.$$

Example 2 – *Evaluating a Line Integral*

Evaluate

$$\int_C (x^2 - y + 3z)\, ds$$

where *C* is the line segment shown in Figure 15.9.

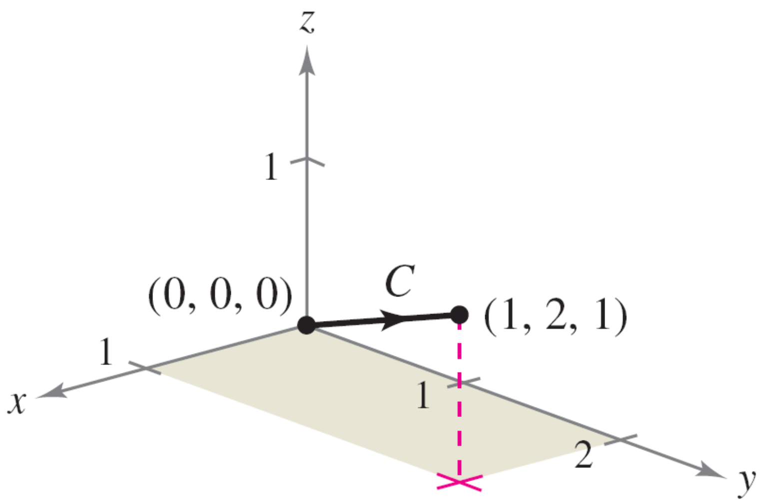


Figure 15.9

Example 2 – *Solution*

Begin by writing a parametric form of the equation of the line segment:

$$x = t, \quad y = 2t, \quad \text{and} \quad z = t, \quad 0 \le t \le 1.$$

Therefore, $x'(t) = 1$, $y'(t) = 2$, and $z'(t) = 1$, which implies that

$$\begin{aligned}\sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} &= \sqrt{1^2 + 2^2 + 1^2} \\ &= \sqrt{6}.\end{aligned}$$

Example 2 – *Solution* cont'd

So, the line integral takes the following form.

$$\int_C (x^2 - y + 3z)\, ds = \int_0^1 (t^2 - 2t + 3t)\sqrt{6}\, dt$$

$$= \sqrt{6}\int_0^1 (t^2 + t)\, dt$$

$$= \sqrt{6}\left[\frac{t^3}{3} + \frac{t^2}{2}\right]_0^1$$

$$= \frac{5\sqrt{6}}{6}$$

Line Integrals

For parametrizations given by $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, it is helpful to remember the form of ds as

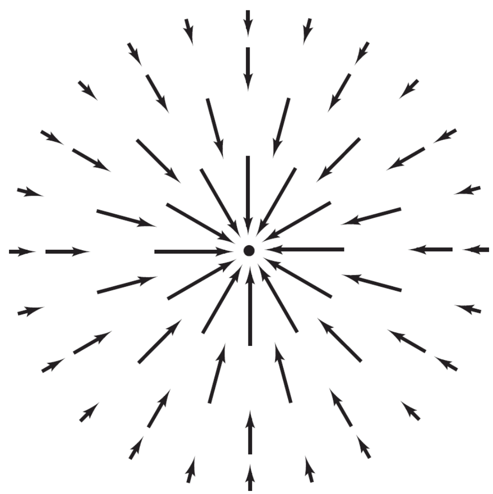
$$ds = \|\mathbf{r}'(t)\|\, dt = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\, dt.$$

Line Integrals of Vector Fields

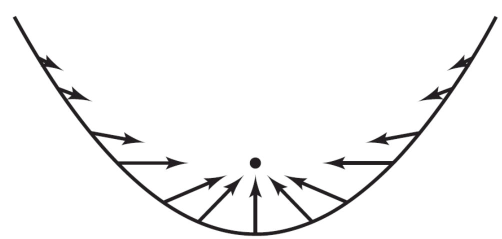
Line Integrals of Vector Fields

One of the most important physical applications of line integrals is that of finding the **work** done on an object moving in a force field.

For example, Figure 15.12 shows an inverse square force field similar to the gravitational field of the sun.



Inverse square force field $\mathbf{F}$

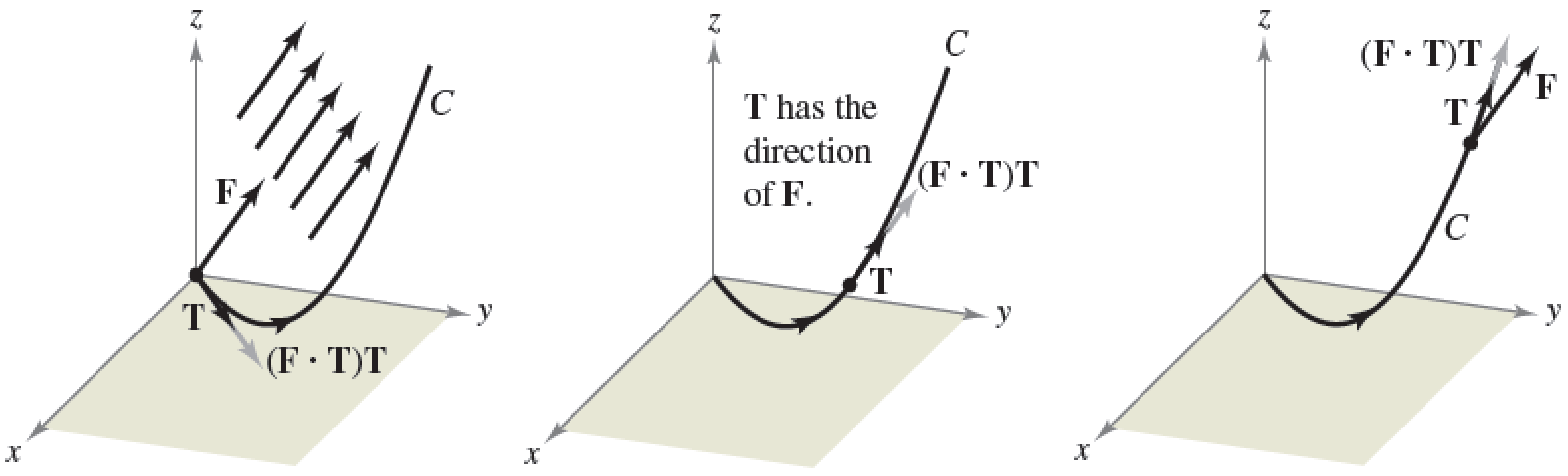


Vectors along a parabolic path in the force field $\mathbf{F}$

Figure 15.12

Line Integrals of Vector Fields

To see how a line integral can be used to find work done in a force field **F**, consider an object moving along a path *C* in the field, as shown in Figure 15.13.



At each point on C, the force in the direction of motion is $(\mathbf{F} \cdot \mathbf{T})\mathbf{T}$.

Figure 15.13

Line Integrals of Vector Fields

To determine the work done by the force, you need consider only that part of the force that is acting in the same direction as that in which the object is moving.

This means that at each point on C, you can consider the projection $\mathbf{F} \cdot \mathbf{T}$ of the force vector $\mathbf{F}$ onto the unit tangent vector $\mathbf{T}$.

On a small subarc of length Δs_i, the increment of work is

$$\Delta W_i = (\text{force})(\text{distance})$$

$$\approx [\mathbf{F}(x_i, y_i, z_i) \cdot \mathbf{T}(x_i, y_i, z_i)]\, \Delta s_i$$

where (x_i, y_i, z_i) is a point in the ith subarc.

Line Integrals of Vector Fields

Consequently, the total work done is given by the integral

$$W = \int_C \mathbf{F}(x, y, z) \cdot \mathbf{T}(x, y, z)\, ds.$$

This line integral appears in other contexts and is the basis of the following definition of the **line integral of a vector field**.

Note in the definition that

$$\begin{aligned}\mathbf{F} \cdot \mathbf{T}\, ds &= \mathbf{F} \cdot \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} \|\mathbf{r}'(t)\|\, dt \\ &= \mathbf{F} \cdot \mathbf{r}'(t)\, dt \\ &= \mathbf{F} \cdot d\mathbf{r}.\end{aligned}$$

Line Integrals of Vector Fields

Definition of the Line Integral of a Vector Field

Let $\mathbf{F}$ be a continuous vector field defined on a smooth curve C given by

$$\mathbf{r}(t), \quad a \le t \le b.$$

The **line integral** of $\mathbf{F}$ on C is given by

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \mathbf{F} \cdot \mathbf{T}\, ds \\ &= \int_a^b \mathbf{F}(x(t), y(t), z(t)) \cdot \mathbf{r}'(t)\, dt.\end{aligned}$$

Example 6 – *Work Done by a Force*

Find the work done by the force field

$$\mathbf{F}(x, y, z) = -\frac{1}{2}x\mathbf{i} - \frac{1}{2}y\mathbf{j} + \frac{1}{4}\mathbf{k} \qquad \text{Force field } \mathbf{F}$$

on a particle as it moves along the helix given by

$$\mathbf{r}(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j} + t\mathbf{k} \qquad \text{Space curve } C$$

from the point (1, 0, 0) to the point (–1, 0, 3π), as shown in Figure 15.14.

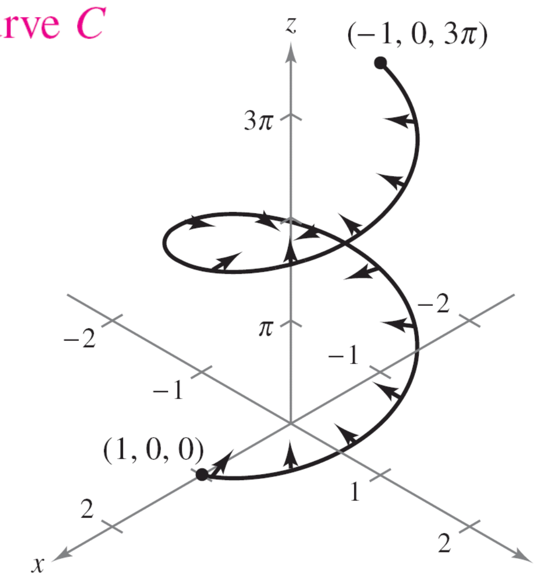


Figure 15.14

Example 6 – *Solution*

Because

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$$

$$= \cos t\,\mathbf{i} + \sin t\,\mathbf{j} + t\,\mathbf{k}$$

it follows that

$$x(t) = \cos t, \qquad y(t) = \sin t, \qquad \text{and} \qquad z(t) = t.$$

So, the force field can be written as

$$\mathbf{F}(x(t), y(t), z(t)) = -\frac{1}{2}\cos t\,\mathbf{i} - \frac{1}{2}\sin t\,\mathbf{j} + \frac{1}{4}\mathbf{k}.$$

Example 6 – *Solution*

cont'd

To find the work done by the force field in moving a particle along the curve C, use the fact that

$$\mathbf{r}'(t) = -\sin t\,\mathbf{i} + \cos t\,\mathbf{j} + \mathbf{k}$$

and write the following.

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} \qquad = \int_a^b \mathbf{F}(x(t), y(t), z(t)) \cdot \mathbf{r}'(t)\, dt$$

$$= \int_0^{3\pi} \left(-\frac{1}{2}\cos t\,\mathbf{i} - \frac{1}{2}\sin t\,\mathbf{j} + \frac{1}{4}\mathbf{k}\right) \cdot (-\sin t\,\mathbf{i} + \cos t\,\mathbf{j} + \mathbf{k})\, dt$$

$$= \int_0^{3\pi} \left(\frac{1}{2}\sin t\cos t - \frac{1}{2}\sin t\cos t + \frac{1}{4}\right) dt$$

$$= \int_0^{3\pi} \frac{1}{4}\, dt \quad = \frac{1}{4}t\Big]_0^{3\pi} \quad = \frac{3\pi}{4}$$

Line Integrals of Vector Fields

For line integrals of vector functions, the orientation of the curve *C* is important.

When the orientation of the curve is reversed, the unit tangent vector **T**(*t*) is changed to –**T**(*t*), and you obtain

$$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_{C} \mathbf{F} \cdot d\mathbf{r}.$$

Line Integrals in Differential Form

Line Integrals in Differential Form

A second commonly used form of line integrals is derived from the vector field notation used in the preceding section.

If **F** is a vector field of the form $\mathbf{F}(x, y) = M\mathbf{i} + N\mathbf{j}$ and C is given by $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$, then $\mathbf{F} \cdot d\mathbf{r}$ is often written as $M\,dx + N\,dy$.

$$\begin{aligned}
\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \mathbf{F} \cdot \frac{d\mathbf{r}}{dt}\,dt \\
&= \int_a^b (M\mathbf{i} + N\mathbf{j}) \cdot (x'(t)\mathbf{i} + y'(t)\mathbf{j})\,dt \\
&= \int_a^b \left(M\frac{dx}{dt} + N\frac{dy}{dt}\right) dt \\
&= \int_C (M\,dx + N\,dy)
\end{aligned}$$

This **differential form** can be extended to three variables.

Line Integrals in Differential Form

The parentheses are often omitted from the differential form, as shown below.

$$\int_C M\,dx + N\,dy$$

In three variables, the differential from is

$$\int_C M\,dx + N\,dy + P\,dz.$$

Example 8 – *Evaluating a Line Integral in Differential Form*

Let *C* be the circle of radius 3 given by

$$\mathbf{r}(t) = 3\cos t\mathbf{i} + 3\sin t\mathbf{j}, \qquad 0 \le t \le 2\pi$$

as shown in Figure 15.17. Evaluate the line integral

$$\int_C y^3\,dx + (x^3 + 3xy^2)\,dy.$$

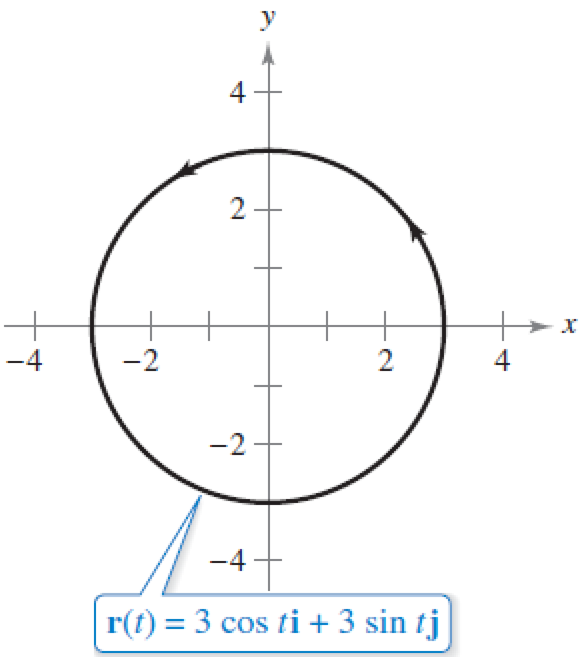


Figure 15.17

Example 8 – *Solution*

Because $x = 3 \cos t$ and $y = 3 \sin t$, you have $dx = -3 \sin t\, dt$ and $dy = 3 \cos t\, dt$. So, the line integral is

$$\int_C M\,dx + N\,dy$$

$$= \int_C y^3\,dx + (x^3 + 3xy^2)\,dy$$

$$= \int_0^{2\pi} [(27 \sin^3 t)(-3 \sin t) + (27 \cos^3 t + 81 \cos t \sin^2 t)(3 \cos t)]\,dt$$

$$= 81 \int_0^{2\pi} (\cos^4 t - \sin^4 t + 3 \cos^2 t \sin^2 t)\,dt$$

Example 8 – *Solution*

cont'd

$$= 81\int_0^{2\pi}\left(\cos^2 t - \sin^2 t + \frac{3}{4}\sin^2 2t\right) dt –$$

$$= 81\int_0^{2\pi}\left[\cos 2t + \frac{3}{4}\left(\frac{1 - \cos 4t}{2}\right)\right] dt$$

$$= 81\left[\frac{\sin 2t}{2} + \frac{3}{8}t - \frac{3\sin 4t}{32}\right]_0^{2\pi}$$

$$= \frac{243\pi}{4}.$$